

Matrices of Linear Transformations

Geometric Algorithms
Lecture 8

Keywords

matrix of a linear transformation

standard basis vectors (standard coordinate vectors)

2D linear transformations

the unit square

one-to-one

onto

Motivating Question

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We know that matrix transformations are linear transformations

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Are there any other kinds of linear transformations?

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Are there any other kinds of linear transformations?

NO

Matrix of a Linear Transformation

Theorem. A transformation T is linear if and only if there is a matrix whose corresponding transformation is T (which "implements" T)

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Linear transformations are **exactly**
matrix transformations

Fundamental Concern

Given a linear transformation T , how do we find the matrix A such that $T(\mathbf{v}) = A\mathbf{v}$?

Thought Experiment

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This is basically linear algebraic battleship

Recall: Calculating $A\mathbf{v}$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} ? \\ ? \\ \vdots \\ ? \end{bmatrix}$$

$$b_1 = a_{11}v_1 + a_{12}v_2 + \cdots + a_{1n}v_n = \sum_{i=1}^n a_{1i}v_i$$

Recall: Matrix-Vector Multiplication

Definition. Given a $(m \times n)$ matrix A with columns $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$, and a vector $\mathbf{v}$ in $\mathbb{R}^n$, we define

$$A\mathbf{v} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1\mathbf{a}_1 + v_2\mathbf{a}_2 + \dots v_n\mathbf{a}_n$$

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$A\mathbf{v}$ is a linear combination of the columns of A with weights given by $\mathbf{v}$

Isolating a_{11}

$$b_1 = a_{11}v_1 + a_{12}v_2 + \dots + a_{1n}v_n = \sum_{i=1}^n a_{1i}v_i$$

The Takeaway

We can learn the first column of the matrix

implementing T by looking at $T \begin{pmatrix} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \end{pmatrix}$

Matrix of a Linear Transformation

Standard Basis

Definition. The *n -dimensional standard basis vectors* (or standard coordinate vectors) are the vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ where

$$\mathbf{e}_i = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ \vdots \\ i-1 \\ i \\ i+1 \\ \vdots \\ n-1 \\ n \end{matrix}$$

Standard Basis

Definition (Alternative). The n -dimensional standard basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ are the columns of the $n \times n$ identity matrix

$$I = [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \dots \quad \mathbf{e}_n]$$

Standard Basis and the Matrix Equation

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] \mathbf{e}_i = \mathbf{a}_i$$

The standard basis vectors gives us a way to
"look into" a matrix

Standard Basis and Vector Coordinates

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \dots + v_n \mathbf{e}_n$$

Column vectors describe how to write a vector as a linear combination of the standard basis

Standard Basis and Linear Transformations

Theorem. For any linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$, the matrix

$$A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \dots \quad T(\mathbf{e}_n)]$$

is the unique matrix such that $T(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v}$ in $\mathbb{R}^n$

Formally

$$T(\mathbf{v}) = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \dots & T(\mathbf{e}_n) \end{bmatrix} \mathbf{v}$$

How To: Matrices of Linear Transformations

Question. Find the matrix which implements the transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$

Solution. Determine the images of standard basis under T . Then write down

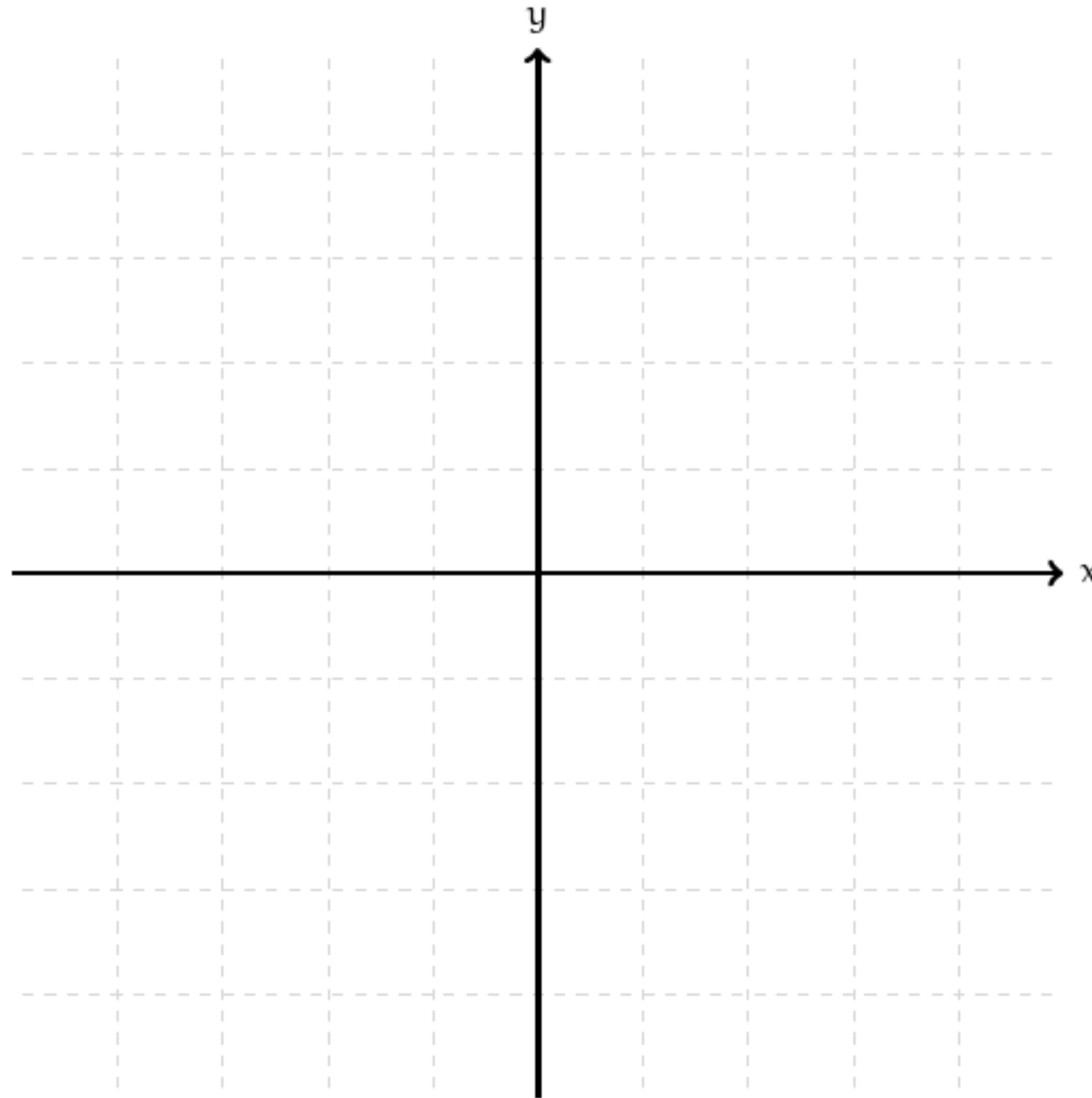
$$[T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \dots \quad T(\mathbf{e}_n)]$$

Geometry of Matrix Transformations

Matrix transformations change the
"shape" of a set of set of
vectors (points).

Example: General Rotation

How does
rotation
affect the
standard
basis?



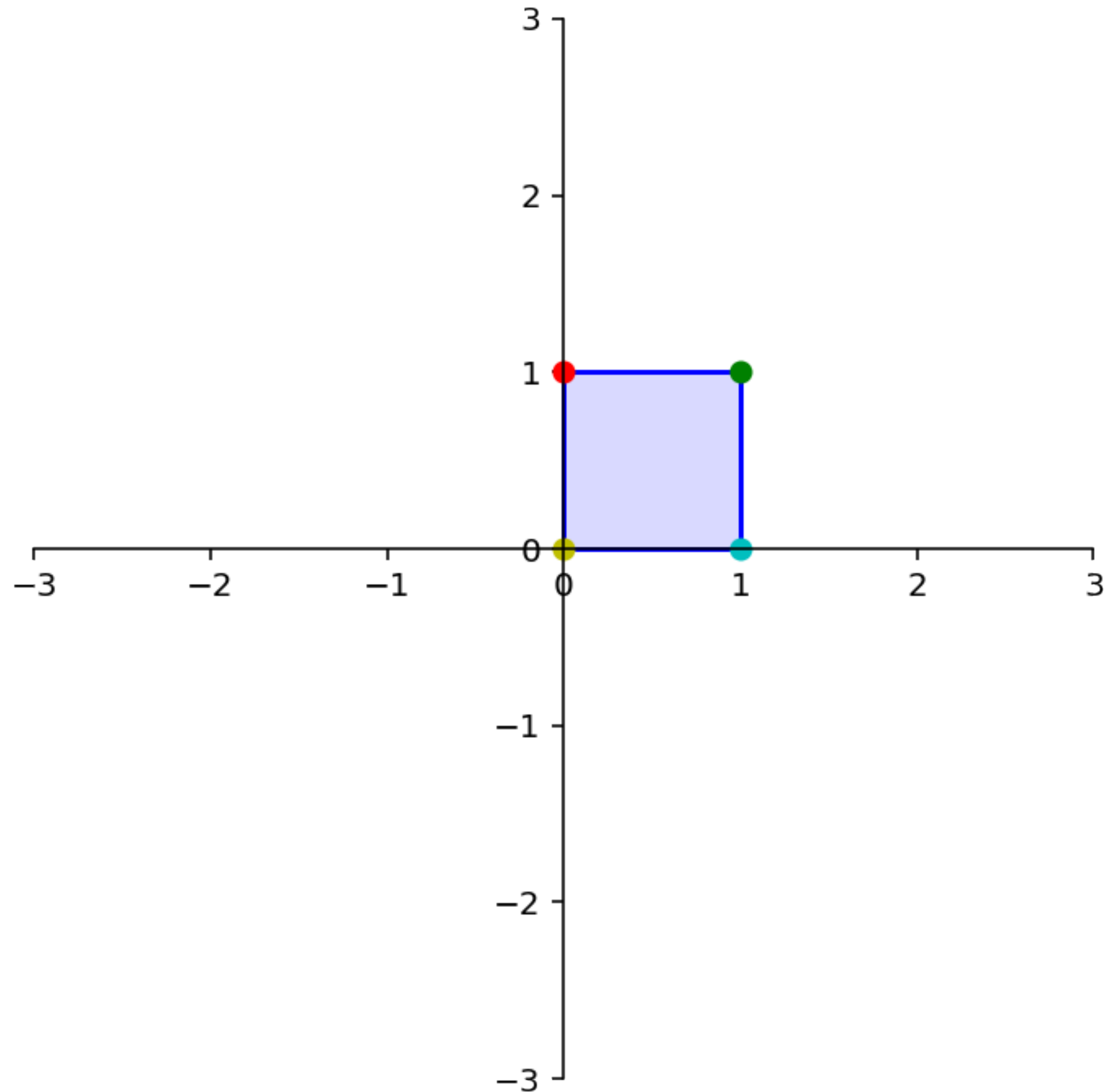
Rotation Matrix

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

The Takeaway: We can figure out the matrix that implements a linear transformations by understanding what it does to the standard basis

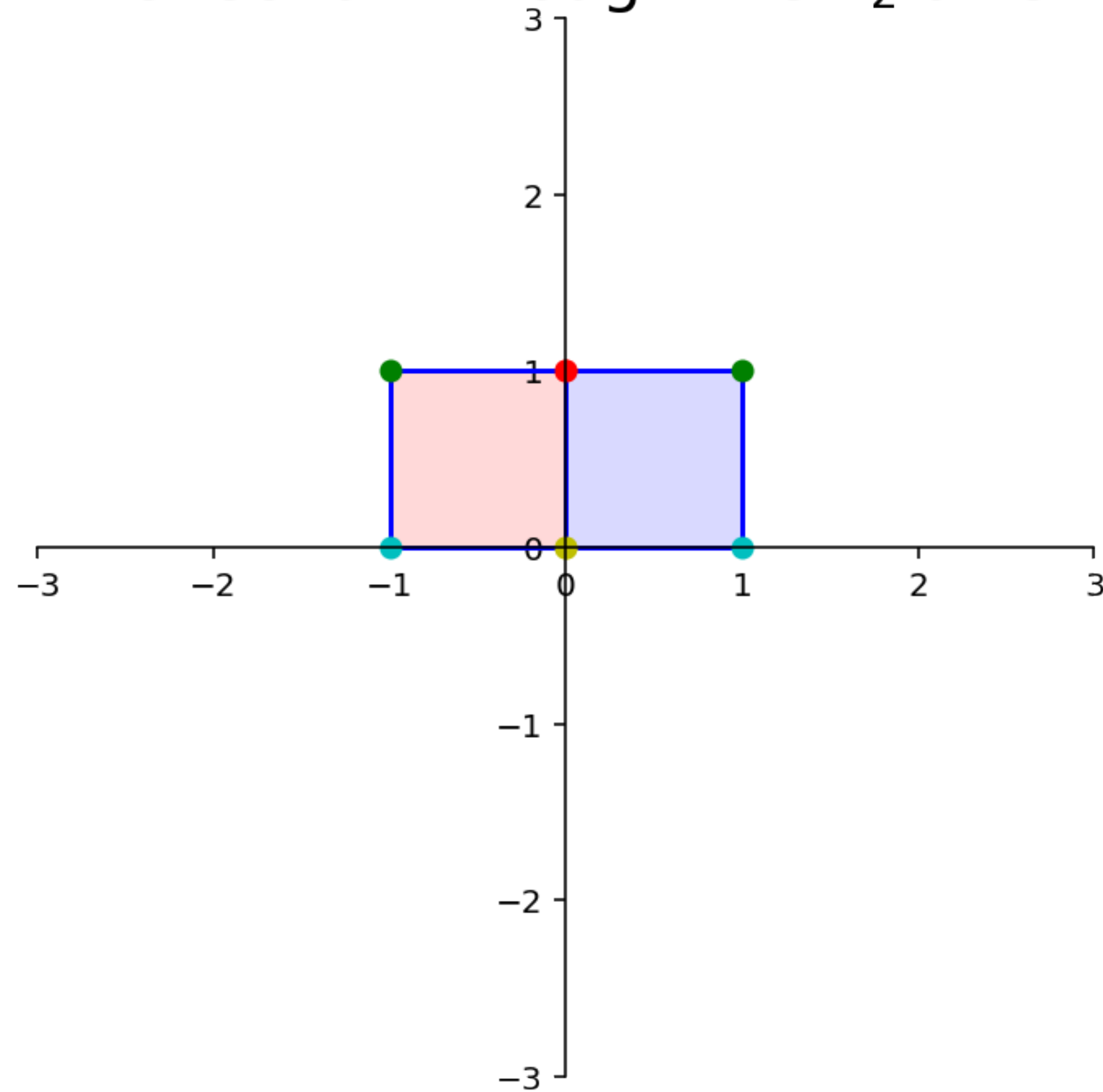
The Unit Square

The *unit square* is the set of points in $\mathbb{R}^2$ enclosed by the points $(0,0)$, $(0,1)$, $(1,0)$, $(1,1)$.



Example: Reflection through the x_2 -axis

Reflection through the x_2 axis



workshop